

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME : DATA ANALYSIS WITH SAS/SPSS/R

PRACTICAL 1

AIM:1. Generating descriptive statistics using summary() or describe() (R).

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The screenshot shows the RStudio IDE with the following components:

- Source:** Contains the R console output. It shows the R version (4.5.2), copyright information, and the installation of the 'dplyr' package. The output includes a warning about Rtools and the successful installation of 'dplyr'.
- Environment:** Lists the loaded data frames: 'AI_Impact_on_Jobs_2030' (3001 obs. of 18 variables), 'arxiv_ai' (10000 obs. of 12 variables), 'category_pivot' (10000 obs. of 919 variables), 'clean_exact' (5 obs. of 3 variables), 'combined_data' (10000 obs. of 2 variables), 'df' (3000 obs. of 19 variables), 'df_clean' (10000 obs. of 4 variables), 'df_part1' (5000 obs. of 2 variables), 'df_part2' (5000 obs. of 2 variables), and 'duplicate_df' (7 obs. of 3 variables).
- Files:** Shows the file explorer with various files and folders, including 'gignore', 'RData', 'Rhistory', 'AI_Impact_on_Jobs_2030.csv', 'analysis.R', 'arxiv_ai.csv', 'prac.6.7.8.9.10', 'prac.6.7.8.9&10.r prog link.R', 'prac.11.12.13.14.15', 'prac.pdfs', 'README.md', and 'RPROG.Rproj'.

The screenshot shows the RStudio IDE with the following components:

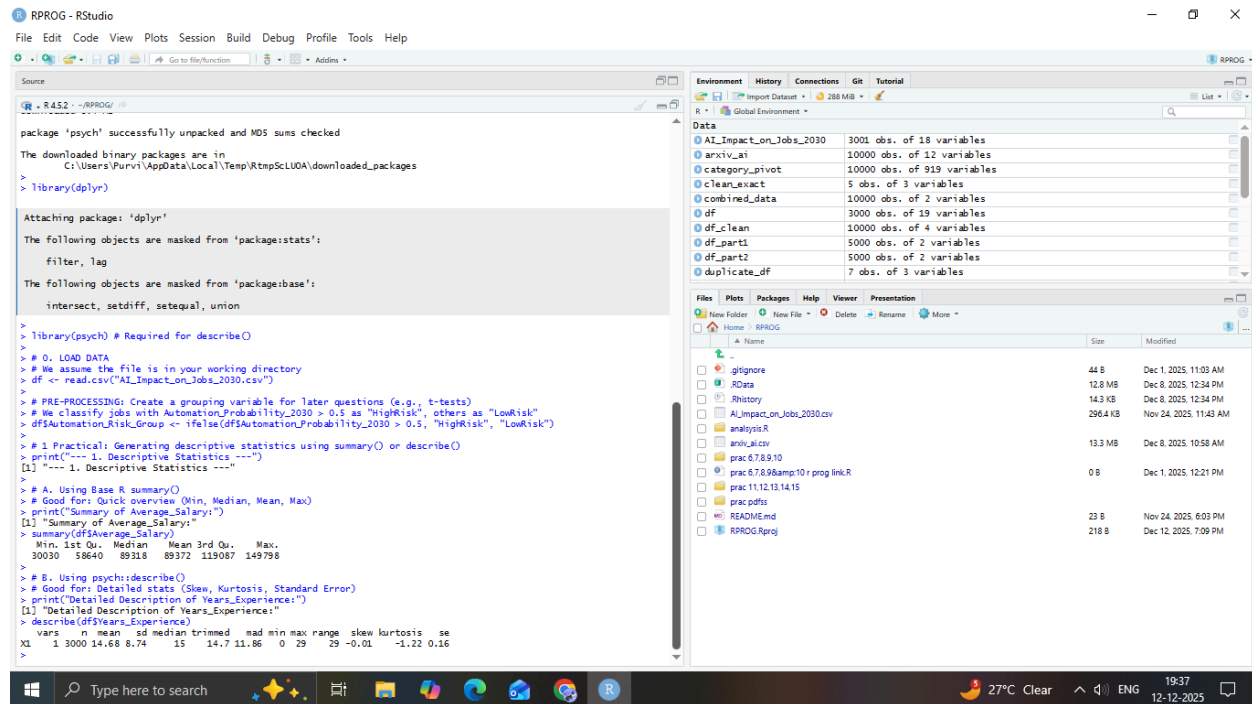
- Source:** Contains the R console output. It shows the installation of the 'psych' package and the loading of the 'dplyr' package. The output includes a warning about Rtools and the successful installation of 'psych'.
- Environment:** Lists the loaded data frames: 'AI_Impact_on_Jobs_2030' (3001 obs. of 18 variables), 'arxiv_ai' (10000 obs. of 12 variables), 'category_pivot' (10000 obs. of 919 variables), 'clean_exact' (5 obs. of 3 variables), 'combined_data' (10000 obs. of 2 variables), 'df' (3000 obs. of 19 variables), 'df_clean' (10000 obs. of 4 variables), 'df_part1' (5000 obs. of 2 variables), 'df_part2' (5000 obs. of 2 variables), and 'duplicate_df' (7 obs. of 3 variables).
- Files:** Shows the file explorer with various files and folders, including 'gignore', 'RData', 'Rhistory', 'AI_Impact_on_Jobs_2030.csv', 'analysis.R', 'arxiv_ai.csv', 'prac.6.7.8.9.10', 'prac.6.7.8.9&10.r prog link.R', 'prac.11.12.13.14.15', 'prac.pdfs', 'README.md', and 'RPROG.Rproj'.

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The screenshot displays the RStudio environment. The source editor on the left contains R code for loading and analyzing data. The environment pane on the right lists the loaded packages and data objects. The console at the bottom shows the output of the executed code.

```
package 'psych' successfully unpacked and MD5 sums checked
The downloaded binary packages are in
C:\Users\Purvi\AppData\Local\Temp\RtmpScLU0A\downloaded_packages
> library(dplyr)

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':
  filter, lag

The following objects are masked from 'package:base':
  intersect, setdiff, setequal, union

> library(psych) # Required for describe()
> # 0. LOAD DATA
> # We assume the file is in your working directory
> df <- read.csv("AI_Impact_on_Jobs_2030.csv")
> # PRE-PROCESSING: Create a grouping variable for later questions (e.g., t-tests)
> # We classify jobs with Automation_Probability_2030 > 0.5 as "HighRisk", others as "LowRisk"
> df$Automation_Risk_Group <- ifelse(df$Automation_Probability_2030 > 0.5, "HighRisk", "LowRisk")
> # 1 Practical: Generating descriptive statistics using summary() or describe()
> print("---- 1. Descriptive Statistics ----")
[1] "---- 1. Descriptive Statistics ----"
> # A. Using Base R summary()
> # Good For: Quick overview (Min, Median, Mean, Max)
> print("Summary of Average_Salary:")
[1] "Summary of Average_Salary:"
> summary(df$Average_Salary)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
30030   58640   89318   89372  119087  149798
> # B. Using psych::describe()
> # Good For: Detailed stats (Skew, Kurtosis, Standard Error)
> print("Detailed Description of Years_Experience:")
[1] "Detailed Description of Years_Experience:"
> describe(df$Years_Experience)
   vars  n mean sd median trimmed  mad min max range skew kurtosis  se
XL    1 3000 14.68 8.74   15   14.7 11.86   0 29  29 -0.01  -1.22 0.16
```

The environment pane shows the following data objects:

Object	Size	Modified
AI_Impact_on_Jobs_2030	3001 obs. of 18 variables	
arxiv_ai	10000 obs. of 12 variables	
category_pivot	10000 obs. of 919 variables	
clean_exact	5 obs. of 3 variables	
combined_data	10000 obs. of 2 variables	
df	3000 obs. of 19 variables	
df_clean	10000 obs. of 4 variables	
df_part1	5000 obs. of 2 variables	
df_part2	5000 obs. of 2 variables	
duplicate_df	7 obs. of 3 variables	

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